2024-Winter Paper 4 – Recent Advances

Contents

1. Q1. Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis. (LAQ)
2. Q2. Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T. (LAQ)
3. Q3a. Newer methods of stone analysis and their significance in management of calculus disease. (SAQ)
4. Q3b. Enlist the steps in female to male gender reassignment surgery. (SAQ)
5. Q3c. Newer generation laser technologies. (SAQ)
6. Q3d. Urolift. (SAQ)
7. Q3e. Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys. (SAQ)
8. Q3f. PARP inhibitors in Prostate cancer. (SAQ)

Q1. Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis.

Type: LAQ (25 marks) (also answered in 2017-Winter Paper 2, Q3b and 2021-Summer Paper 4, Q3d)

Read as: recent trends in the evaluation, diagnosis and management of IC/BPS (interstitial cystitis/bladder pain syndrome).

Definition: how it has changed

- **NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987):** research criteria; missed about 60% of patients, so the definition was broadened.
- **AUA (American Urological Association), current:** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or other identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (>6 months) bladder pain with at least one other urinary symptom, after confusable diseases are excluded.
- **ICD-11 (International Classification of Diseases, 11th revision):** "chronic primary bladder pain syndrome".
- **Trend: splitting the syndrome.** "IC" is reserved for **Hunner lesion (HL) disease**, a true inflammatory disease; BPS without HL is a symptom complex. Both sit within UCPPS (urological chronic pelvic pain syndrome).

Pathophysiology (current view)

- Multifactorial: leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation; **central sensitisation**; stress-axis dysfunction; small-fibre neuropathy. No infection proven.
- HL disease shows dense lymphoplasmacytic infiltrate; non-HL bladders are often histologically normal.
- **Associated:** IBS (irritable bowel syndrome, >30%), fibromyalgia, vulvodynia, endometriosis; tender pelvic floor in about 80%.

Recent trends in evaluation (AUA 2022)

- **Diagnosis of exclusion plus localisation:** tests aim to find the pain generator.
- **History:** pain worse on filling, relieved by voiding; "painful urgency" without urge incontinence, unlike OAB (overactive bladder). **Ask about ketamine.**
- **Examination: six-site pelvic-floor palpation** (anterior and posterior levator ani, obturator internus, each side); vulval swab test; DRE (digital rectal examination) in men.
- **Basic tests:** PVR (post-void residual), urinalysis, culture; cytology if risk factors; frequency–volume chart.
- **Scores:** O'Leary–Sant ICSI/ICPI (Interstitial Cystitis Symptom/Problem Index), GUPI (Genitourinary Pain Index), pain VAS (visual analogue scale); **score pain and urinary symptoms separately.**
- **Body map:** widespread pain = 3 or more of 12 regions besides the pelvis.
- **Office cystoscopy if over 50 or HL suspected** (AUA 2022 change): HL is a stellate red area with radiating vessels, found only by looking.
- **Local anaesthetic challenge** (lidocaine ± bupivacaine) localises pain to the bladder and selects intravesical therapy.
- **Dropped:** potassium sensitivity test (neither sensitive nor specific); routine hydrodistension (glomerulations non-specific); routine urodynamics.
- **Biopsy** only to exclude CIS (carcinoma in situ).

ESSIC classification (cystoscopy with hydrodistension + biopsy, e.g. "3C"):

Cystoscopy	Biopsy
X not done; 1 normal; 2 glomerulations; 3 Hunner lesion	X not done; A normal; B inconclusive; C positive (inflammation, mastocytosis, fibrosis)

Exclude: UTI (urinary tract infection) including tuberculosis, CIS, stones, radiation or cyclophosphamide cystitis, ketamine cystitis, OAB, diverticulum, endometriosis.

Phenotype-based diagnosis (key trend)

Phenotype	Identified by	Treatment
Hunner lesion	Cystoscopy	Fulguration and/or triamcinolone; ciclosporin A if refractory
Anaesthetic-responsive bladder pain	Local anaesthetic challenge	Intravesical therapy
Pelvic-floor tenderness	Pelvic-floor examination	Myofascial physiotherapy
Widespread (centralised) pain	Body map, associated pain conditions	Systemic: amitriptyline, gabapentinoids, CBT (cognitive behavioural therapy), pain team

- **MAPP (Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain) reanalysis, 2023:** hydroxyzine/PPS (pentosan polysulfate) helped localised pain; amitriptyline helped widespread pain.

Recent trends in management (AUA 2022)

- **Tiers abandoned:** the six-tier ladder is replaced by grouped options and shared decisions; multimodal treatment allowed early; major surgery last. Stop treatments that fail.

Group	Options
Behavioural	Education, avoid trigger foods, stress management, flare plan
Physical	**Myofascial pelvic-floor therapy** (Standard); avoid Kegel exercises
Oral	**Amitriptyline** 10 mg at night, up to about 50 mg; hydroxyzine; cimetidine; **PPS 100 mg three times daily** (pigmentary maculopathy: baseline and periodic eye checks)

Group	Options
Intravesical	**DMSO (dimethyl sulfoxide) 50%**; heparin; alkalinised lidocaine; also hyaluronic acid, chondroitin
Procedures	**HL: fulguration and/or triamcinolone** (75–90% improve); short low-pressure hydrodistension; **onabotulinumtoxinA 100 U into the trigone** (must accept self-catheterisation); SNM (sacral neuromodulation)
Refractory HL	**Oral ciclosporin A** (watch kidney function and blood pressure)
Last resort (<10%)	Cystoplasty or diversion ± cystectomy

- **Not offered:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral steroids.

Recent advances

- Urinary biomarkers and functional MRI (magnetic resonance imaging) for phenotyping (research); phenotype-stratified trials.
- **Also know:** tanezumab (anti-NGF [nerve growth factor]) not adopted; liposomal intravesical botulinum toxin no better than placebo; pudendal neuromodulation.

Diagram to draw: cystoscopic Hunner lesion with radiating vessels, and the four-phenotype treatment chart.

Key points to remember:

- AUA definition: bladder pain with LUTS >6 weeks; ESSIC >6 months.
- AUA 2022: no tiers; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test.
- HL: fulguration or triamcinolone; ciclosporin if refractory.
- PPS causes maculopathy; antibiotics and BCG not offered.

Q2. Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T.

Type: LAQ (25 marks)

Introduction

- A **cancer vaccine** is a therapeutic agent that trains the patient's own T cells to recognise **tumour antigens**. It treats existing cancer; it does not prevent it.
- **BCG (bacille Calmette–Guérin)**, a tuberculosis vaccine, was among the first immune treatments for any cancer. It is non-specific. Cancer vaccines aim for a **specific** anticancer response.

Rationale in urology

- **Immune editing (three Es):** Elimination → Equilibrium → **Escape**. A clinical cancer has escaped by losing antigens, losing MHC (major histocompatibility complex), or recruiting Tregs (regulatory T cells). Vaccines try to restart the response.
- **Kidney:** rare spontaneous regression of metastases after nephrectomy; responses to IL-2 (interleukin-2) and interferon.
- **Prostate:** slow growth gives time to build a response; **organ-specific antigens** (PSA [prostate-specific antigen], PAP [prostatic acid phosphatase]).
- **Bladder:** BCG success; high mutation load gives many neoantigens.
- Low toxicity; possible lasting memory.

Mechanism of action

- Two components: **antigen** (overexpressed in tumour) + **adjuvant** (a danger signal, e.g. GM-CSF [granulocyte-macrophage colony-stimulating factor], which activates DCs [dendritic cells]).

```
Intradermal antigen + adjuvant
  → immature DC takes up antigen; adjuvant matures it
  → DC migrates to lymph node
  → peptide on MHC I/II = signal 1; B7 costimulation = signal 2
  → CD4 Th1 help (IL-2, interferon-γ) + CD8 cytotoxic T cells
  → T cells traffic to tumour → apoptosis of antigen-bearing cells
  → memory T cells
```

- Without signal 2 the T cell becomes tolerant, hence the adjuvant.
- **Prime–boost:** first dose primes, later doses boost.
- **Delayed effect:** survival may improve without tumour shrinkage or PSA fall.
- **Resistance:** PD-L1 (programmed death ligand-1) on tumour, Tregs, MDSCs (myeloid-derived suppressor cells), MHC loss. This is the basis for combining vaccines with checkpoint inhibitors.

Types (platforms)

Platform	Example
Whole-cell (autologous or allogeneic)	Reniale (RCC [renal cell carcinoma]); GVAX (prostate)
Peptide/protein	IMA901
Dendritic-cell (ex vivo loaded)	Sipuleucel-T; AGS-003 (rocapuldencel-T)
Viral vector (poxvirus)	PROSTVAC-VF; TroVax (MVA-5T4)

Platform	Example
Heat-shock protein–peptide	Vitespen (HSPPC-96)
DNA/mRNA, personalised neoantigen	Phase I–II trials

Vaccines by organ

Renal cell carcinoma

- **IMA901:** 10 tumour-associated peptides + GM-CSF, low-dose cyclophosphamide to deplete Tregs; HLA-A*02+ (human leucocyte antigen) metastatic RCC with sunitinib. No OS (overall survival) benefit.**
- **AGS-003 (rocapuldencel-T):** autologous DCs loaded with tumour RNA, with sunitinib. **Terminated for lack of efficacy.**
- **TroVax** with standard first-line therapy: no OS benefit. **Vitespen** (adjuvant): no overall benefit.
- **Reniale** (adjuvant autologous tumour-cell vaccine): claimed benefit, not adopted.
- Checkpoint inhibitors, not vaccines, are today's RCC immunotherapy.

Prostate cancer

- **Sipuleucel-T:** the **only approved** vaccine (below).
- **PROSTVAC-VF:** vaccinia prime, fowlpox boost, encoding PSA + TRICOM (three costimulatory molecules). Phase III **closed for futility**.
- **GVAX:** allogeneic whole-cell lines secreting GM-CSF. Phase III trials **failed**.
- Also: DCVAC/PCa (phase III negative); DNA vaccines against PAP (trials).

Bladder cancer

- **BCG:** intravesical live attenuated mycobacteria; standard for high-risk NMIBC (non-muscle-invasive bladder cancer).
- **Gene and viral therapy: nadofaragene firadenovec** (adenovirus carrying IFN-α2b gene; FDA [Food and Drug Administration] approved December 2022 for BCG-unresponsive CIS [carcinoma in situ]); **cretostimogene** (oncolytic adenovirus with GM-CSF gene).
- **N-803:** IL-15 superagonist given with BCG.
- Peptide and neoantigen vaccines: experimental only.

Why most vaccines failed

- Tolerance to self-antigens; immunosuppressive tumour environment; dysfunctional patient DCs; bulky late disease; progression endpoints miss delayed effects.
- **Pattern: only sipuleucel-T succeeded in phase III.**

Note on Sipuleucel-T

- **Nature:** autologous cellular immunotherapy: the patient's **CD54+ APCs (antigen-presenting cells)** activated with **PA2024**, a fusion protein of **PAP + GM-CSF**.
- **Manufacture and dosing:**
  1. Leukapheresis collects PBMCs (peripheral blood mononuclear cells).
  2. Cells incubated about 36 h with PA2024 at a central facility.
  3. Infused intravenously about 3 days after collection.
  4. **Three doses at 2-week intervals** (one month course).
- **Mechanism:** primes PAP-specific CD4 and CD8 T cells; later infusions boost.
- **IMPACT trial (Kantoff, NEJM 2010):** 512 men, 2:1 versus placebo, asymptomatic or minimally symptomatic mCRPC (metastatic castration-resistant prostate cancer).
  - **OS 25.8 vs 21.7 months (HR [hazard ratio] 0.78)**, gain 4.1 months, despite placebo crossover.

    - **No change in time to progression or PSA.**
- **FDA approval April 2010:** first therapeutic vaccine for a solid tumour.
- **Indication:** asymptomatic or minimally symptomatic mCRPC, good performance status, **no visceral metastases, no opioid need**; usually before chemotherapy.
- **Toxicity:** mild infusion reactions (fever, chills, fatigue); premedicate with paracetamol and antihistamine.
- **Limitations:** cost, logistics, no response marker; not available in India.

Diagram to draw: sipuleucel-T flow: leukapheresis → PA2024 loading → reinfusion → T-cell priming.

> **Key points to remember:**
> - Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
> - IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
> - Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months; no PSA or progression effect.
> - BCG is the original urological immunotherapy.

Q3a. Newer methods of stone analysis and their significance in management of calculus disease.

Type: SAQ (10 marks)

Why analyse stones

- Analyse a stone at least once in every former (AUA [American Urological Association], EAU [European Association of Urology]); repeat if stones recur on treatment.
- Composition points to the cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection.

Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**, so it cannot separate COM (calcium oxalate monohydrate, whewellite) from COD (dihydrate, weddellite), or apatite from brushite.
- High error rates; misses rare stones (2,8-DHA [2,8-dihydroxyadenine], drugs); needs a large sample.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; small sample; identifies proteins and drugs
XRD (X-ray diffraction)	Crystal lattice diffracts X-rays in a unique pattern	Gold standard for crystals; misses amorphous material
Polarising microscopy	Crystal optics (birefringence)	Cheap; experienced observer
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section + FTIR	Subtypes (e.g. type Ia COM low urine volume, Ic primary hyperoxaluria)
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface structure, elements	Research
Micro-CT (micro-computed tomography)	3-D non-destructive structure	Research
Raman spectroscopy	Scattered light spectrum	Research

In vivo prediction

- **DECT (dual-energy computed tomography):** two energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid stones** with near-perfect accuracy.
- **Hounsfield units:** >1000 HU predicts SWL (shock wave lithotripsy) failure.
- Endoscopic recognition (lost when stones are dusted).

Significance in management

- **Uric acid:** oral dissolution by urinary alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture, antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy.
- **Brushite:** often converts from calcium oxalate; avoid over-alkalinisation.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency, treat with allopurinol. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: FTIR spectrum outline and COM versus COD crystal shapes.

Key points to remember:

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

Q3b. Enlist the steps in female to male gender reassignment surgery.

Type: SAQ (10 marks)

Preconditions

- Gender dysphoria diagnosed per **WPATH (World Professional Association for Transgender Health)** by an MDT (multidisciplinary team).
- **Testosterone:** chapter cites **at least 12 months** before genital surgery (clitoral growth plateaus at about 2 years). WPATH SOC8 (Standards of Care 8, 2022) relaxed this to about 6 months.
- **Urological screening:** uroflowmetry, PVR (post-void residual), renal ultrasound; treat LUTD (lower urinary tract dysfunction) first.

Stage 1: removal of female structures

1. **Subcutaneous mastectomy** (usually first).
2. **Hysterectomy + bilateral salpingo-oophorectomy**, preferably laparoscopic; remove whole cervix ± upper third of vagina.

Stage 2: genitoperineal transformation

3. **Vaginectomy** (needed if the urethra is lengthened):
    - Hydrodissection; circumferential introital incision.
    - Rectovaginal plane bluntly; urethrovaginal plane sharply, **no diathermy near the sphincter**.
    - Paravaginal haemostasis; levators closed in midline.
4. **Urethral lengthening, fixed (perineal) part:** vestibular mucosa and labia minora tubularised over 18 Fr; covered with bulbospongiosus.
5. **Phallus construction**, either:
    - **Metoidioplasty:** release clitoral chordee and urethral plate; lengthen urethra to the glans (preputial or labial flap ± buccal graft). Gives a 3–8 cm phallus with erection and sensation.
    - **Phalloplasty: RFFF (radial forearm free flap)** or ALT (anterolateral thigh) flap with **tube-in-tube urethra**; microvascular anastomosis; **clitoral nerve coapted** to a flap nerve.
6. **Urethral anastomosis** of phallic to fixed part; neomeatus at the tip.
7. **Scrotoplasty:** labia majora flaps rotated 180° in front (Ghent type). Monsplasty and perineal closure.
8. **SPC (suprapubic catheter)** + urethral catheter; urethral catheter out at 14–21 days.

Stage 3: prostheses

9. **Testicular prostheses** (17 mL) at 6 months or more.
10. **Erectile prosthesis** after 1 year, once protective sensation returns; single cylinder fixed to pubic bone (dedicated devices, e.g. ZSI 475 FTM [female-to-male]).

Complications

- **Fistula** (phallic base), **stricture** (fixed–phallic anastomosis), hair growth, dribbling, prosthesis infection or erosion, vaginectomy bleeding or rectal/bladder injury.

Diagram to draw: three parts of the neourethra (native, fixed, phallic).

Key points to remember:

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Q3c. Newer generation laser technologies.

Type: SAQ (10 marks)

Background

- **Ho:YAG (holmium:yttrium-aluminium-garnet), 2120 nm:** standard for stones and enucleation; water chromophore; 0.4 mm penetration; **photothermal**.
- Newer lasers aim at **less retropulsion, finer dust, faster work**.

1. High-power holmium (60–120 W)

- High frequency allows **dusting** (0.2–0.3 J × 40–80 Hz) and "popcorning"; long-pulse modes reduce retropulsion.

2. Pulse modulation (Moses technology and similar)

- **Moses effect:** the start of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone.
- **Moses technology** splits each pulse in two: the first opens the channel, the second hits the target.
- Less retropulsion; may shorten **HoLEP (holmium laser enucleation of prostate)**.

3. TFL (thulium fibre laser), 1940 nm

- **Diode-pumped silica fibre**, at the **water absorption peak** (about 4× holmium absorption); penetration about 0.2 mm.
- **Low peak power, long pulses, very high frequency (above 2000 Hz).**
- **Thin fibres** (down to about 150 µm): more irrigation and scope deflection.
- **Stones:** finer dust, minimal retropulsion. RCTs (randomised controlled trials): better than **low-power holmium**; **equal** to 120 W pulse-modulated holmium.
- **Prostate:** ThuFLEP (thulium fibre laser enucleation).
- Compact and quiet.

4. Thulium:YAG (2013 nm)

- Continuous wave: ThuVARP (vaporesection), ThuVEP (vapoenucleation), **ThuLEP (laser enucleation)**; carbonisation a drawback.
- **Pulsed Tm:YAG:** newer pulsed solid-state version for lithotripsy and enucleation.

Comparison

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Water absorption	High	About 4× higher
Peak power	High	Low
Retropulsion	More	Minimal
Fibre	≥200 µm	≥150 µm

Safety (applies to all)

- **Heat follows power, not laser type:** ≤10 W in the ureter, ≤25 W in the kidney; cool irrigation and access sheath (damage above 43 °C). Eyewear for all.

Others

- **Also know:** KTP/LBO (potassium titanyl phosphate/lithium triborate) 532 nm vaporisation; diode lasers; suction sheaths.

Diagram to draw: water absorption curve marking 1940 and 2120 nm.

> **Key points to remember:**
> - TFL 1940 nm: water peak, low peak power, minimal retropulsion.
> - Moses: two-part pulse through a vapour channel.
> - TFL beats low-power, equals high-power pulse-modulated holmium.
> - Heat depends on power: ≤10 W in the ureter.

Q3d. Urolift.

Type: SAQ (10 marks)

Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).

Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**. Permanent implants pull the lateral lobes apart and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**; extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

Implant and mechanism

- Spring-loaded implant with T-shaped ends:
  - **Capsular tab:** nitinol.
  - **Urethral end:** stainless steel (resists encrustation).
  - **Connector:** polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. MRI (magnetic resonance imaging)-conditional.

Indications

- **AUA (American Urological Association): prostate 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe, men wanting ejaculation preserved.
- Men wishing to stop medication.
- **Avoid:** obstructing median lobe, large gland, active infection.

Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, from bladder neck to apex, avoiding the neurovascular bundles.
4. **Keep implants >1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

Evidence

	L.I.F.T. (2013)	BPH6 (2015)
Design	PUL vs sham, 2:1, 19 centres	PUL vs TURP (transurethral resection of prostate), 10 European centres
Result	IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s; no new sexual dysfunction	Composite end point met 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6%
Durability	13.6% surgical retreatment at 5 years	TURP better IPSS, Qmax, PVR (post-void residual)

Complications

- Transient dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device malfunction.

- **Higher retreatment than TURP.**

Diagram to draw: axial prostate with implants pulling lateral lobes to the capsule.

> **Key points to remember:**
> - 30–80 mL, no median lobe; local anaesthesia.
> - Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
> - TURP wins on IPSS and flow.

Q3e. Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys.

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 3, Q3f)

"Preemptive" is read as pre-implantation (procurement) biopsy.

Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- Still better survival than dialysis (Ojo, 2001).

Categories

- **ECD (expanded criteria donor), UNOS (United Network for Organ Sharing):**
  - Age **60 or over**, or
  - Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.
  - Graft-loss risk about 1.7 times that of a standard donor.
- **KDPI (Kidney Donor Profile Index) >85%** replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Others:** DCD (donation after circulatory death); donor AKI (acute kidney injury); HCV-positive donors; older or hypertensive living donors.

Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components, each 0–3: total 0–12.**

Component	0	1	2	3
Glomerular global sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial and arteriolar narrowing	None	Wall thickness < lumen diameter	Wall = or slightly > lumen	Wall far > lumen, severe narrowing or occlusion

Allocation by score

Total	Grade	Use
0–3	Mild	**Single** kidney transplant
4–6	Moderate	**Dual** kidney transplant
7–12	Severe	**Discard**

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

Optimising marginal kidneys

- Short cold ischaemia; **HMP (hypothermic machine perfusion)** reduces DGF; old-for-old matching; informed consent.

Diagram to draw: Remuzzi grid with four compartments scored 0–3.

> **Key points to remember:**
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
> - Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
> - 0–3 single, 4–6 dual, ≥7 discard.

Q3f. PARP inhibitors in Prostate cancer.

Type: SAQ (10 marks)

Introduction

- **PARP (poly-ADP-ribose polymerase)** repairs single-strand DNA breaks. **PARPIs (PARP inhibitors):** olaparib, rucaparib, niraparib, talazoparib.

Mechanism: synthetic lethality

- **Catalytic inhibition** and **PARP trapping** on DNA leave single-strand breaks unrepaired.
- At replication these become **double-strand breaks**, normally repaired by **HRR (homologous recombination repair)**.
- Tumours lacking HRR (e.g. **BRCA2** loss) die; normal cells survive.
- **AR (androgen receptor)–PARP synergy:** AR maintains HRR genes, so AR blockade adds repair deficiency: basis for PARPI + ARPI (androgen receptor pathway inhibitor).

Who benefits: testing

- **HRR alterations in mCRPC (metastatic castration-resistant prostate cancer): up to 20–25%.** Genes: **BRCA2 (largest benefit)**, BRCA1, ATM (weak), PALB2, CHEK2, CDK12.
- **Germline testing** in all metastatic disease, strong family history or Ashkenazi ancestry; **somatic testing** on tissue or ctDNA (circulating tumour DNA), where clonal haematopoiesis can mislead.

Key trials

Trial	Setting	Result
PROfound	Olaparib vs enzalutamide/abiraterone after ARPI	BRCA/ATM cohort: imaging PFS (progression-free survival) **7.4 vs 3.6 months (HR [hazard ratio] 0.34)**; OS (overall survival) 19.1 vs 14.7 months in the final analysis (chapter quotes the interim 18.5 vs 15.1)
TRITON3	Rucaparib vs physician's choice	BRCA: PFS 11.2 vs 6.4 months (HR 0.50); no ATM benefit
PROpel	Olaparib + abiraterone, first-line mCRPC	OS 42.1 vs 34.7 months (P = 0.054, not significant); greatest in BRCA
MAGNITUDE	Niraparib + abiraterone, HRR-altered	BRCA rPFS (radiographic PFS) 19.5 vs 10.9 months (HR 0.55); no benefit without HRR alteration
TALAPRO-2	Talazoparib + enzalutamide, first line	All-comers rPFS not reached vs 21.9 months (HR 0.63); largest in BRCA

- **Also know:** TOPARP-A/B, TRITON2, GALAHAD (phase II).

Approvals (FDA [Food and Drug Administration])

- **Olaparib monotherapy:** HRR-mutated mCRPC after ARPI (2020).
- **Rucaparib:** BRCA-mutated mCRPC.
- **Olaparib or niraparib + abiraterone:** BRCA-mutated; **talazoparib + enzalutamide:** HRR-mutated (both 2023).

Toxicity

- **Anaemia** (commonest grade 3–4; monitor counts), fatigue, nausea; rare **MDS/AML (myelodysplastic syndrome/acute myeloid leukaemia)**, pulmonary embolism.

Recent advances

- **mCSPC (metastatic castration-sensitive prostate cancer)** trials: AMPLITUDE, TALAPRO-3; ATR inhibitors for ATM loss.

Diagram to draw: synthetic lethality: PARP block + BRCA loss → cell death.

> **Key points to remember:**
> - Synthetic lethality; BRCA2 gains most, ATM least.
> - PROfound: olaparib, PFS 7.4 vs 3.6 months.
> - Combinations: PROpel, MAGNITUDE, TALAPRO-2.

Rapid Revision Sheet – 2024-Winter Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis. (LAQ)

- AUA definition: bladder pain with LUTS >6 weeks; ESSIC >6 months.
- AUA 2022: no tiers; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test.
- HL: fulguration or triamcinolone; ciclosporin if refractory.
- PPS causes maculopathy; antibiotics and BCG not offered.

Q2. Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T. (LAQ)

- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months; no PSA or progression effect.
- BCG is the original urological immunotherapy.

Q3a. Newer methods of stone analysis and their significance in management of calculus disease. (SAQ)

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

Q3b. Enlist the steps in female to male gender reassignment surgery. (SAQ)

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Q3c. Newer generation laser technologies. (SAQ)

- TFL 1940 nm: water peak, low peak power, minimal retropulsion.
- Moses: two-part pulse through a vapour channel.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- Heat depends on power: ≤10 W in the ureter.

Q3d. Urolift. (SAQ)

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
- TURP wins on IPSS and flow.

Q3e. Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys. (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- 0–3 single, 4–6 dual, ≥7 discard.

Q3f. PARP inhibitors in Prostate cancer. (SAQ)

- Synthetic lethality; BRCA2 gains most, ATM least.

- PROfound: olaparib, PFS 7.4 vs 3.6 months.
- Combinations: PROpel, MAGNITUDE, TALAPRO-2.